

Matrix Theory: Lecture 11

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Outline

- Schur's lemma
- Normal Matrices
- Hermitian Forms
- Congruent Matrices
- Diagonalization



Schur's Lemma

Theorem 13.6

Every square matrix A can be written as

$$A = URU^H,$$

where U is a unitary matrix and R is an upper triangular matrix. (The decomposition is not unique.)

For every linear operator there exists an orthonormal basis in which its matrix is upper triangular.



Normal Matrices

What matrices can be diagonalized by a unitary matrix?

Def. We say that a matrix is normal if $A^H A = A A^H$.

(Examples: Hermitian, skew-Hermetian, unitary.)

Theorem 13.7

Let A be a square matrix.

- ① A is normal iff A^H is normal.
- ② A is normal $\Leftrightarrow UAU^H$ is normal for any unitary matrix U .
- ③ A is normal and invertible $\Rightarrow A^{-1}$ is normal.
- ④ If A is a block matrix

$$A = \begin{pmatrix} B & O \\ O & C \end{pmatrix}$$

with square blocks B, C then A is normal $\Leftrightarrow B$ and C are normal.

- ⑤ If A is a triangular matrix then A is normal if and only if it is a diagonal matrix.

Normal Matrices

Theorem 13.8

The following conditions for a square matrix A are equivalent:

- 1 A is a normal matrix.
- 2 A can be diagonalized by a unitary matrix U , i.e.
 $U^{-1}AU = \text{diag}(\lambda_1, \dots, \lambda_n)$.
- 3 A has an orthonormal basis consisting of eigenvectors.
- 4 $A = UDU^H$ for some unitary matrix U and some diagonal matrix D .
- 5 The absolute values of the non-zero eigenvalues (considered in decreasing order) coincide with the singular values.



Normal Matrices

Corollary 13.9

Let A be a normal matrix and let $\lambda_1, \dots, \lambda_n$ be its eigenvalues. Then

- ① A is Hermitian iff all λ_k are real.
- ② A is skew-Hermitian iff $\operatorname{Re}(\lambda_k) = 0$ for every λ_k . In particular A is skew-Hermitian iff iA is Hermitian.
- ③ A is unitary iff $|\lambda_k| = 1$ for every λ_k .



Hermitian Forms

Def. Let V be a complex vector space. A **Hermitian form** on V is a function of form

$$f(X) = \sum_{i,j} a_{ij} \bar{x}_i x_j, = X^H A X$$

where X is a vector with the (complex!) coordinates x_i w.r.t. to some basis and $A = (a_{ij})$ is a **Hermitian** matrix.

Note that $f(X)$ is a real number since

$$X^H A X = (X, AX) = (A^H X, X) = (AX, X) = \overline{(X, AX)} = \overline{X^H A X}$$

If X, A are real and A is **symmetric** then

$$f(X) = \sum_{i,j} a_{ij} x_i x_j = X^T A X,$$

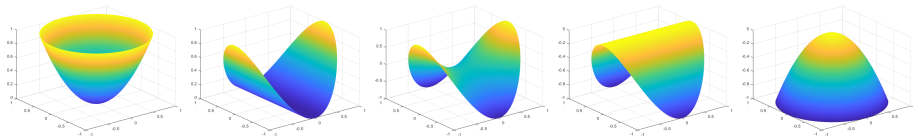
is called a **Quadratic form**.



Hermitian Forms

The Hermitian (in real case the quadratic) form and the corresponding matrix A are called

- **positive definite** if $f(X) > 0$ for $X \neq O$;
- **positive semidefinite** if $f(X) \geq 0$ for all X ;
- **indefinite** if $f(X)$ attains both negative and positive values.
- **negative semidefinite** if $f(X) \leq 0$ for all X ;
- **negative definite** if $f(X) < 0$ for $X \neq O$;



Congruent Matrices

Def. Two Hermitian matrices A and B are **congruent** if $B = S^H A S$ for some invertible matrix S .

If $X = S\hat{X}$ then

Linear operator:	Hermitian form:
$Y = AX$	$f(X) = X^H A X$
$\hat{A} = S^{-1} A S$	$\hat{A} = S^H A S$
A and $\hat{A}$ are similar	A and $\hat{A}$ are congruent



Diagonalization of Hermitian Forms

Theorem 14.3

For every Hermitian form $f(X) = X^H A X$ one can choose a basis such that in this basis

$$f(\hat{X}) = \lambda_1 |\hat{x}_1|^2 + \cdots + \lambda_n |\hat{x}_n|^2.$$

Here λ_i are eigenvalues of the original matrix A . In particular,

- $f(X)$ is positive definite iff all eigenvalues are positive;
- $f(X)$ is positive semidefinite iff all eigenvalues are non-negative;
- $f(X)$ is negative definite iff all eigenvalues are negative;
- $f(X)$ is negative semidefinite iff all eigenvalues are non-positive;
- $f(X)$ is indefinite if it has both positive and negative eigenvalues.

